DIPANKAR R. BAISYA

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PROFESSIONAL SUMMARY

Applied Scientist specializing in LLMs, RAG, Multi-Agent AI Systems, and production-grade generative AI. Demonstrated success in fraud detection using classical deep learning and modern multimodal LLM approaches. Experienced in fine-tuning, scaling, and architecting cloud-native AI solutions with proven business impact from internship to senior scientist level.

EXPERIENCE

Applied Scientist II | *Amazon, Seattle* | Winter 2022 – Present

- Primary ML POC for amazon.mx since 2023, maintaining all ML models for MX retail with cross-functional collaboration; developed LLM-based fraud detection using pre-trained embeddings and multimodal RAG systems, reducing bad debt by 7%
- Developed multi-agent risk-based trading algorithms with Model Context Protocol (MCP) orchestrating 60+ real-time financial APIs, achieving 20% improvement in output consistency through agent specialization
- Led LLM fine-tuning using LoRA, Red Teaming, DPO, and GRPO for fraud detection and security; optimized model inference through Quantization, knowledge Distillation, Distributed training, and Mixed Precision (FP16/BF16)
- Promoted from Level 4 to Level 5 (Summer 2023) for technical excellence and business impact

Applied Scientist Intern | *Amazon, Seattle* | Summer 2019 & 2020

- Implemented model interpretation for decision tree-based models and deep learning architectures (LSTM, GRU) to improve customer experience
- Applied Auto-encoder and bidirectional recurrent neural networks (BLSTM, BGRU) for real-time fraud detection during transactions

Graduate Research Assistant | *University of California, Riverside* | Fall 2016 – Winter 2022

- Applied deep learning approaches to biological sequence prediction problems, including CRISPR-Cas9 guide RNA prediction and histone modification prediction
- Published research in high-impact journals including Nature Communications and Bioinformatics

AI PROJECTS

RiskNavigator AI: Multi-Agent Financial Advisor (2025)

GitHub | Live Demo | Blog

- Architected production-grade 6-agent hierarchical AI system implementing Agent-to-Agent (A2A) communication protocol and Model Context Protocol (MCP) to orchestrate 60+ real-time financial APIs for comprehensive automated financial analysis and risk assessment
- Researched multi-agent vs. monolithic LLM architectures through comparative analysis, demonstrating 20% improvement in output consistency and reduced hallucination through agent specialization
- Deployed serverless cloud architecture with auto-scaling infrastructure (0-10 instances) achieving 99.9% uptime, delivering multi-dimensional risk assessment (market, liquidity, company-specific) with automated strategy generation and PDF report export

Multimodal RAG System (2025)

GitHub | Live Demo

- Developed multimodal RAG system integrating video, audio, and text data with preprocessing workflows (OpenCV, Whisper) and BridgeTowerEmbeddings pipeline using LanceDB vector database for efficient cross-modal storage and retrieval
- Deployed production system on HuggingFace Spaces using Pixtral vision-language model for multimodal conversation generation

Agentic RAG System (2025)

GitHub | Live Demo

- Developed Agentic RAG system using LangGraph orchestrating multi-step workflows with Qwen2.5-Coder-32B-Instruct, integrating specialized tools (Wikipedia, Arxiv, Tavily web search, mathematical operations) and Supabase vector database for semantic similarity search
- Evaluated system performance using **GAIA** benchmark and deployed production system on HuggingFace Spaces

Build and Deploy GPT-2 from Scratch (2025)

GitHub | Live Demo | Blog

- Built complete GPT architecture from scratch (124M-1,558M parameters) with tokenization, multi-head attention, layer normalization, and residual connections; validated against OpenAI weights and fine-tuned for spam classification (97% accuracy) and instruction-following with DPO alignment
- Deployed production chatbot on HuggingFace Spaces with controlled text generation (temperature scaling, top-k sampling) and comprehensive technical documentation

LLM & AI SKILLS

- **Large Language Models:** GPT (OpenAI), Llama, Mistral, Claude, Qwen, Gemini, BERT, Sentence-BERT, Pixtral
- **Frameworks:** HuggingFace Transformers, LangChain, LangGraph, PyTorch, TensorFlow, Google Agent Development Kit (ADK),
- **Distributed Training:** FSDP, DeepSpeed, Hugging Face Accelerate, Torchtitan, Mixed Precision Training (FP16/BF16)
- **Profiling:** PyTorch Profiler, Nvidia NSight
- **LLM Techniques:** Fine-tuning, Prompt Engineering, RAG, RLHF, PEFT, LoRA, QLoRA, Red Teaming, Agent Orchestration, DPO, GRPO
- **Multi-Agent Systems:** Agent-to-Agent (A2A) Communication, State-Based Coordination, Hierarchical Agent Architecture, Model Context Protocol (MCP)
- **Multimodal:** Vision-Language Models (VLMs), OpenAI CLIP, Whisper, BridgeTower, BLIP
- **Vector Databases:** LanceDB, FAISS, ChromaDB, Supabase
- **Cloud & MLOps:** Amazon AWS, Lambda AI, Modal, Google Cloud Run, Vertex AI, Docker, Cloud Build, FastAPI, Serverless Architecture,

EDUCATION

Ph.D. in Computer Science | *University of California, Riverside* | Winter 2022

B.Sc. in Computer Science and Engineering | *Bangladesh University of Engineering & Technology* | February 2013

CERTIFICATIONS & TRAINING

- **Kaggle & Google:** 5 - Day AI Agents Intensive Course with Google - Advanced techniques in building and deploying production-grade AI agent systems
- **HuggingFace:** Foundations of Agents - LangGraph, SmolAgent, LangChain, CrewAI
- **Coursera:** Generative AI with LLMs, CNNs, Structuring ML Projects, Improving Deep Neural Networks
- **Maven:** Scratch to Scale: Large Scale Training in Modern World - DDP, ZeRO Optimization, Pipeline and Tensor Parallelism, MoE, Mixed Precision training (FP16/BF16)

SELECTED PUBLICATIONS

- Baisya, D. R., Ramesh, A., Schwartz, C., Lonardi, S., & Wheeldon, I. "Genome-wide functional screens enable the prediction of high activity CRISPR-Cas9 and-Cas12a guides in Yarrowia lipolytica." Nature Communications (2022)
- Baisya, Dipankar Ranjan, and Stefano Lonardi. "Prediction Of Histone Post-Translational Modifications Using Deep Learning." Bioinformatics (2020)
- Baisya, D. R., Wu, Yibbing., Raghebi, Zohreh. "BOTSpot: Fast Automatic Detection of BOT Attack" AMLC (2024), Submitted